

Mehedi Hassan, Ph.D.

INSTRUCTOR, DEPARTMENT OF ELECTRICAL ENGINEERING AND COMPUTER SCIENCE

University of Arkansas, Fayetteville, AR 72701, USA.

✉ mehedi1798@gmail.com 🎓 Google Scholar 🔗 LinkedIn

RESEARCH INTERESTS

My research lies at the intersection of cybersecurity, machine learning, and complex networked systems. I focus on developing ML methods for detecting anomalous behaviors in dynamic networks, including suspicious communication patterns, anomalous links, and evolving network threats. My long-term research goal is to develop scalable, adaptive, and interpretable ML approaches that strengthen the security, resilience, and integrity of critical digital infrastructure, improve cyber risk detection, and support timely responses to emerging cybersecurity challenges.

EDUCATION

Ph.D. in Computer Science

UNIVERSITY OF LOUISIANA AT LAFAYETTE

Jan 2019-Dec 2024

Lafayette, Louisiana, USA

- Dissertation: *Detecting Anomalous Links in Dynamically Evolving Complex Systems*
- Advisor: Dr. Mehmet Engin Tozal

M.Sc. in Computer Science

UNIVERSITY OF LOUISIANA AT LAFAYETTE

Jan 2019-Dec 2021

Lafayette, Louisiana, USA

- Research focus: machine learning, network security, and anomaly detection
- Advisor: Dr. Mehmet Engin Tozal

B.Sc. in Computer Science and Engineering

ISLAMIC UNIVERSITY OF TECHNOLOGY

Jan 2013-Nov 2016

Gazipur, Bangladesh

- Thesis: *Traffic Prioritization Using Packet Aggregation in Saturated WSN*
- Advisor: Dr. Abu Raihan Mostofa Kamal

PUBLICATIONS

Journal Articles

- [J5] N. Heidarikohol, M.E. Tozal, **Mehedi Hassan**. "Impact of Vaccination on COVID-19 Spread Patterns in the US States." In *Applied Operations and Analytics*, 2026.
- [J4] **Mehedi Hassan**, M.E. Tozal, V. Swarup, S. Noel, R. Gottumukkala, V. Raghavan. "Detecting Anomalous Links in Networked Systems Exhibiting Scale-Free-Like Behaviors." In *IEEE Systems Journal*, 2025. **(Impact Factor: 4.4; Q1)**.
- [J3] **Mehedi Hassan**, M.E. Tozal, V. Swarup, S. Noel, R. Gottumukkala, V. Raghavan. "Detecting Anomalous Communication Behaviors in Dynamically Evolving Networked Systems." In *IEEE Transactions on Information Forensics and Security*, 2025. **(Impact Factor: 8.0; Q1)**.
- [J2] **Mehedi Hassan**, M.E. Haque, M.E. Tozal. "Efficacy of the Confinement Policies on the COVID-19 Spread Dynamics in the Early Period of the Pandemic." In *Intelligent Data Analysis*, vol. 27, no. 3, pp. 855-884, 2023.
- [J1] **Mehedi Hassan**, M.E. Haque, M.E. Tozal, V.R. Raghavan, R. Agarwal. "Intrusion Detection using Payload Embeddings." In *IEEE Access*, vol. 10, pp. 4015-4030, 2021. **(Impact Factor: 3.6; Q1)**.

Conference Papers

- [C5] T. Tahmida, S.T. Pasha, **Mehedi Hassan**, A. Islam, M.A. Amin. "Integrating Supervised and Self-Supervised Models for an Enhanced PCOS Detection: A Data-Driven Approach with Machine Learning Perspective." In *Proceedings of the 6th International Conference on Artificial Intelligence, Robotics and Control*, Savannah, GA, USA, 2025, pp. 499-503.
- [C4] N. Halder, T.Y. Bristy, M. Arshad, H. Ratul, M.Z. Alam, **Mehedi Hassan**, A. Islam, M.A. Amin. "Performance Analysis of Machine Learning Models for Detecting Anomalous Traffic in Internet of Medical Things." In *Proceedings of the 6th International Conference on Artificial Intelligence, Robotics and Control*, Savannah, GA, USA, 2025, pp. 482-487.
- [C3] **Mehedi Hassan**, M.E. Tozal, V. Swarup, S. Noel, R. Gottumukkala, V. Raghavan. "Anomalous Link Detection in Dynamically Evolving Scale-Free-Like Networked Systems." In *Proceedings of the 18th Annual IEEE International Systems Conference*, 2024, pp. 1-8.

- [C2] S. Roy, M. Ashaduzzaman, **Mehedi Hassan**, A.R. Chowdhury. “BlockChain for IoT Security and Management: Current Prospects, Challenges and Future Directions.” In *Proceedings of the 5th International Conference on Networking, Systems and Security*, 2018.
- [C1] M.Z.I. Alam, A. Rahman, **Mehedi Hassan**, H. Mahmud, M.K. Hasan. “EXPRESSit: Interactive Messaging via Gestures.” In *Proceedings of the 19th International Conference on Computer and Information Technology*, 2016.

Book Chapters

- [B2] M.U. Islam, M.M. Mottalib, **Mehedi Hassan**, Z.I. Alam, S.M. Zobaed, M.F. Rabby. “The Past, Present, and Prospective Future of XAI: A Comprehensive Review.” In *Explainable Artificial Intelligence for Cyber Security: Next Generation Artificial Intelligence*, Springer Nature, pp. 1-30, 2022.
- [B1] S.M. Zobaed, **Mehedi Hassan**, M.U. Islam, M.E. Haque. “Deep Learning in IoT-Based Healthcare Applications.” In *Deep Learning for Internet of Things Infrastructure*, CRC Press, pp. 183-200, 2021.

Theses

- [T2] **Mehedi Hassan**. “Detecting Anomalous Links in Dynamically Evolving Complex Systems.” Ph.D. dissertation, *University of Louisiana at Lafayette*, 2024.
- [T1] **Mehedi Hassan**. “Traffic Prioritization Using Packet Aggregation in Saturated WSN.” B.Sc. thesis, *Islamic University of Technology*, 2016.

Under Review, Revision, and Preparation

- [O2] **Mehedi Hassan**, M.E. Tozal, V. Swarup, S. Noel. “Enhancing Graph Attention Networks for Anomalous Link Detection in Complex Systems.” Under revision at *IEEE Transactions on Artificial Intelligence*.
- [O1] **Mehedi Hassan**, M.E. Tozal. “Understanding the Variations in the Early COVID-19 Spread Patterns in the United States.” Under revision at *Informatics for Health and Social Care*.
- [P2] **Mehedi Hassan**, S. Ahmad. “Temporal Graph Neural Networks for Emerging Cyber Threat Detection in Dynamic Networked Systems.” (In preparation).
- [P1] **Mehedi Hassan**, M.U. Islam. “Explainable AI for Graph-Based Anomaly Detection in Complex Systems.” (In preparation).

SPONSORED RESEARCH PROJECTS (CONTRIBUTOR)

List of funded research projects to which I contributed as a Graduate Research Assistant during my graduate studies.

- | | |
|---|---------------------|
| [SP2] <i>Predictive Link Analyses in Computer Network Domains</i> | May-Dec 2023 |
| Funding Agency: The MITRE Corporation (Award # MITRE-1161269-A1) | |
| [SP1] <i>Partnership for COVID-19 Vaccination for Underserved Populations in Louisiana (VAX-UP Louisiana)</i> | Aug-Dec 2021 |
| Funding Agency: Louisiana Department of Health (Contract # 2000595087) | |

RESEARCH APPOINTMENTS

- | | |
|--|---------------------|
| [R4] Graduate Research Assistant | May-Dec 2023 |
| EIGEN LAB, UNIVERSITY OF LOUISIANA AT LAFAYETTE | |
| <i>Lafayette, Louisiana, USA</i> | |
| Advisor: <i>Dr. Mehmet Engin Tozal</i> | |
| My research focused on machine learning-based anomaly detection in dynamically evolving networked systems under a MITRE-sponsored research project. In collaboration with The MITRE Corporation, I developed machine learning and graph-based methods for anomalous communication behavior detection, suspicious link detection, and cybersecurity analytics. This work contributed to peer-reviewed publications in <i>IEEE SysCon</i> , <i>IEEE Transactions on Information Forensics and Security</i> , and <i>IEEE Systems Journal</i> [C3, J3, J4]. | |
| [R3] Graduate Research Assistant | May-Aug 2022 |
| EIGEN LAB, UNIVERSITY OF LOUISIANA AT LAFAYETTE | |
| <i>Lafayette, Louisiana, USA</i> | |
| Advisor: <i>Dr. Mehmet Engin Tozal</i> | |
| My research focused on studying COVID-19 spread dynamics during the early period of the pandemic. I analyzed how confinement policies affected temporal spread patterns across countries and contributed to machine learning-based analysis of pandemic response measures. This work resulted in a journal publication in <i>Intelligent Data Analysis</i> [J2]. | |

- [R2] **Graduate Research Assistant** **Aug-Dec 2021**
 INFORMATICS RESEARCH INSTITUTE, UNIVERSITY OF LOUISIANA AT LAFAYETTE *Lafayette, Louisiana, USA*
Advisor: Dr. Mehmet Engin Tozal
 My research focused on analyzing COVID-19 vaccination and pandemic spread patterns in the United States. I contributed to data-driven modeling of state-level pandemic trajectories, vaccination effects, and regional variations in spread dynamics. This work contributed to a journal article in *Applied Operations and Analytics* and an ongoing manuscript in *Informatics for Health and Social Care* [J5, O1].
- [R1] **Graduate Research Assistant** **May-Aug 2020**
 EIGEN LAB, UNIVERSITY OF LOUISIANA AT LAFAYETTE *Lafayette, Louisiana, USA*
Advisor: Dr. Mehmet Engin Tozal
 My research focused on intrusion detection using representation learning for network payloads. I contributed to developing payload embedding-based methods for detecting malicious network activity and improving data-driven intrusion detection. This work resulted in a journal publication in *IEEE Access* [J1].

RESEARCH COLLABORATIONS

- [RC1] **Collaborator** **Aug 2024–Present**
 CENTER FOR COMPUTATIONAL & DATA SCIENCES (CCDS), INDEPENDENT UNIVERSITY, BANGLADESH
 - Conduct collaborative research on machine learning, cybersecurity analytics, anomalous behavior detection, and complex networked systems.
 - Mentor student researchers and support research activities involving data analysis, model development, experimentation, and scholarly publication.

ACADEMIC APPOINTMENTS

- [A4] **Instructor** **Aug 2025–Present**
 DEPARTMENT OF EECS, UNIVERSITY OF ARKANSAS *Fayetteville, Arkansas, USA*
 - Designed and instructed undergraduate computer science and data science courses, including *Programming Foundations I*, *Introduction to Object-Oriented Programming*, and *Intro to AI for Security*.
 - Developed lectures, labs, assignments, exams, grading rubrics, and classroom activities for programming, machine learning, and cybersecurity-related topics.
- [A3] **Visiting Instructor** **Aug 2024-Jul 2025**
 DEPARTMENT OF IT, GEORGIA SOUTHERN UNIVERSITY *Statesboro, Georgia, USA*
 - Designed and instructed undergraduate courses on *Computer Programming I and II*, *Information Systems Security*, *IT Infrastructure*, *Web Programming*, and *Discrete Mathematics*.
 - Developed a graduate-level *Seminar in IT* course integrating research-driven case studies and methodologies to strengthen analytical and problem-solving skills in IT.
 - Led the undergraduate internship course by setting academic goals, monitoring student progress, and helping students connect classroom learning with industry experience.
- [A2] **Graduate Teaching Assistant** **Jan 2019-Jul 2024**
 SCHOOL OF COMPUTING AND INFORMATICS, UNIVERSITY OF LOUISIANA AT LAFAYETTE *Lafayette, Louisiana, USA*
 - Designed and instructed an undergraduate *Information Assurance/Security* course.
 - Graded and mentored undergraduate and graduate students in courses including *Information Assurance and Security*, *Information Management*, *Systematic Methods in Informatics*, and *Data Analysis and Visualization*.
- [A1] **Lecturer** **Jan 2017-Dec 2018**
 DEPARTMENT OF CSE, GREEN UNIVERSITY OF BANGLADESH *Dhaka, Bangladesh*
 - Instructed theory courses and conducted laboratory sessions covering *Programming Languages*, *Database Management Systems*, *Computer Networks*, *Software Engineering*, and *Data Communication*.
 - Led the application process for IEB accreditation for the Department of Computer Science.

PRESENTATIONS & TALKS

Paper Presentations

- **AIRC 2025**, “Integrating Supervised and Self-Supervised Models for an Enhanced PCOS Detection: A Data-Driven Approach with Machine Learning Perspective”, May, 2025.
- **AIRC 2025**, “Performance Analysis of Machine Learning Models for Detecting Anomalous Traffic in Internet of Medical Things”, May, 2025.
- **IEEE SysCon 2024**, “Anomalous Link Detection in Dynamically Evolving Scale-Free-Like Networked Systems”, April, 2024.
- **NSysS 2018**, “BlockChain for IoT Security and Management: Current Prospects, Challenges and Future Directions”, December, 2018.
- **ICCIT 2016**, “EXPRESSit: Interactive Messaging via Gestures”, December, 2016.

Invited Research Talks

- “Payload2Vec: Learning Network Payload Representations for Intrusion Detection”, Research Seminar, University of Louisiana at Lafayette, February, 2020.

STUDENT MENTORSHIP

- **Niloofar Heidarikohol** (M.S., University of Louisiana at Lafayette): *COVID-19 vaccination impact analysis using ML*. [J5]
- **Joseph G. Bordelon** (B.Sc., University of Louisiana at Lafayette): *ML for COVID-19 spread pattern analysis*. [O1]
- **Tasmia Tahmida Jidney** (B.Sc., Independent University Bangladesh): *PCOS detection using machine learning*. [C5]
- **Nabarun Halder** (B.Sc., Independent University Bangladesh): *Anomalous traffic detection in IoMT environments*. [C4]

PROFESSIONAL SERVICES

Technical Program Committee

- 6th International Conference on Artificial Intelligence, Robotics, and Control (AIRC) 2025

Journal Reviewer

- Informatics for Health and Social Care (IHSC) 2026-Present
- IEEE Journal of Biomedical and Health Informatics (JBHI) 2023-Present
- The Journal of Supercomputing (J. Supercomput.) 2023-Present
- IEEE Access 2021-Present
- IEEE Transactions on Network and Service Management (TNSM) 2020-Present
- International Journal of Computers and Applications (IJCA) 2020-Present

Conference Reviewer

- International Conference on Computer Sciences (IHCONCS) 2026
- 6th International Conference on Artificial Intelligence, Robotics, and Control (AIRC) 2025
- 24th International Conference on Machine Learning and Applications (ICMLA) 2025

AWARDS

- **Bangladesh-Sweden Trust Fund Scholarship**, *Bangladesh-Sweden Trust Fund*, 2019.
- **Admission Test Excellency Scholarship**, *Islamic University of Technology*, 2013.
- **Education Board Scholarship**, *Government of Bangladesh*, 2009 & 2011.

TECHNICAL SKILLS AND TOOLS

- **Generative AI**: ChatGPT, Gamma, Gemini, Claude, NotebookLM, Perplexity.
- **Programming**: Python, C, C++, Java, SQL, MATLAB, JavaScript, PHP, HTML, CSS.
- **Research**: LaTeX, PyTorch, TensorFlow, Keras, NetworkX, DGL, GeoPandas, Plotly, Bokeh, Git, GitHub, Linux.
- **Cybersecurity**: Scapy, Wireshark, Nmap, Aircrack-ng, John The Ripper, Hashcat, HoneyPot, Snort, tcpdump.
- **Teaching & Productivity**: Google Colab, OnlineGDB, ZyBooks, Blackboard, WebAssign, J&B Learning, NetLab.

PROFESSIONAL MEMBERSHIPS

- Member, Institute of Electrical and Electronics Engineers (IEEE) Jan 2024-Present